

Link: <https://studio.edgeimpulse.com/public/990916/live>

Lab 6 submission:

1. The raw inputs were accX, accY, and accZ, a three-axis accelerometer time series.
2. The neural network classifier had 33 input features, some of the names are accX RMS, accX Peak 1 Height, accY Peak 3 height, accX Peak 1 Freq, and accY RMS. It processed spectral features.
3. There were two output classes, nominal and off.
4. The features passed into the anomaly detector were accX RMS, accX Peak 1 Height, accY Spectral power 2.0-5.0 hz, and accZ Spectral Power 2.0-5.0hz.

```
Anomaly prediction: 16.015026
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 49 ms, inference 583 us, anomaly 525 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 15.958893
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 49 ms, inference 593 us, anomaly 557 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 15.955334
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 49 ms, inference 564 us, anomaly 555 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.013556
Starting inferencing in 2 seconds...
```

5. |
6. The Edge Impulse model successfully produced inference results on the Arduino. The classifier output should not always be fully trusted because the output depends on the diversity of the training data and what it knows. A lot of conditions can cause it to not be reliable. For example, environmental noise and where the sensor is placed.

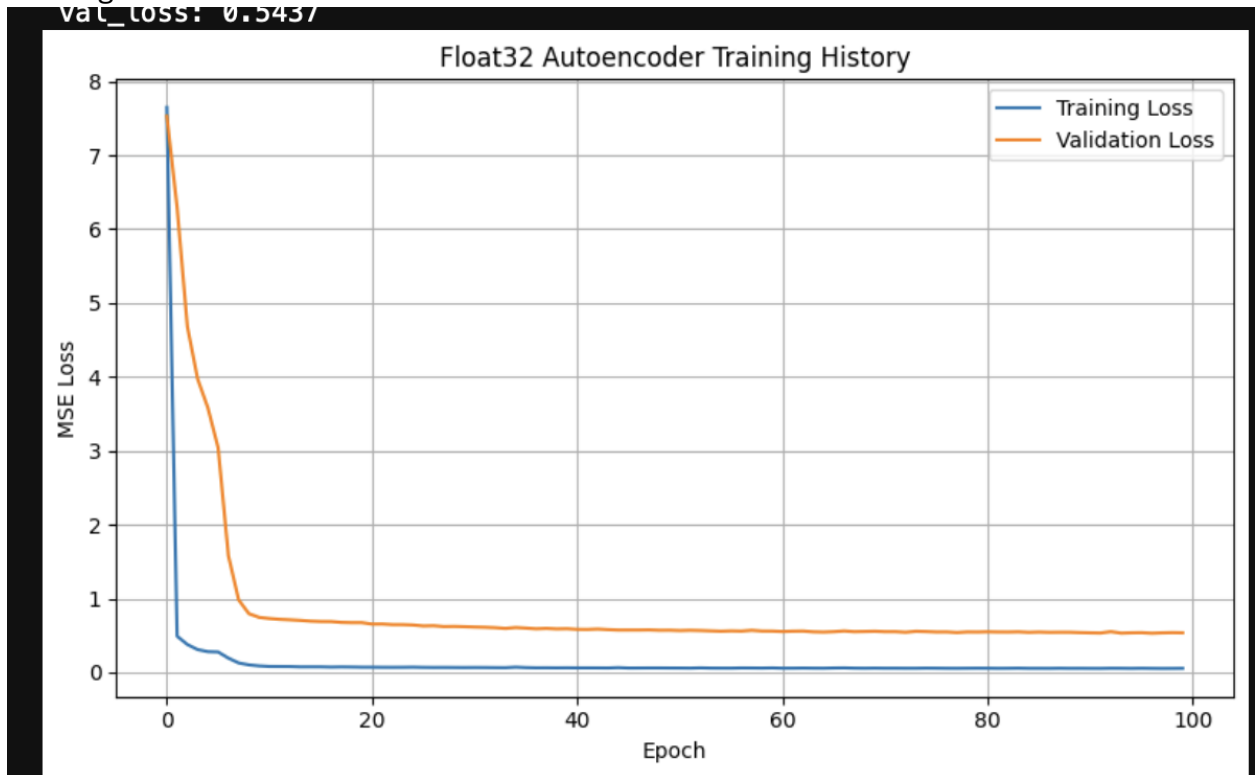
Part 2:

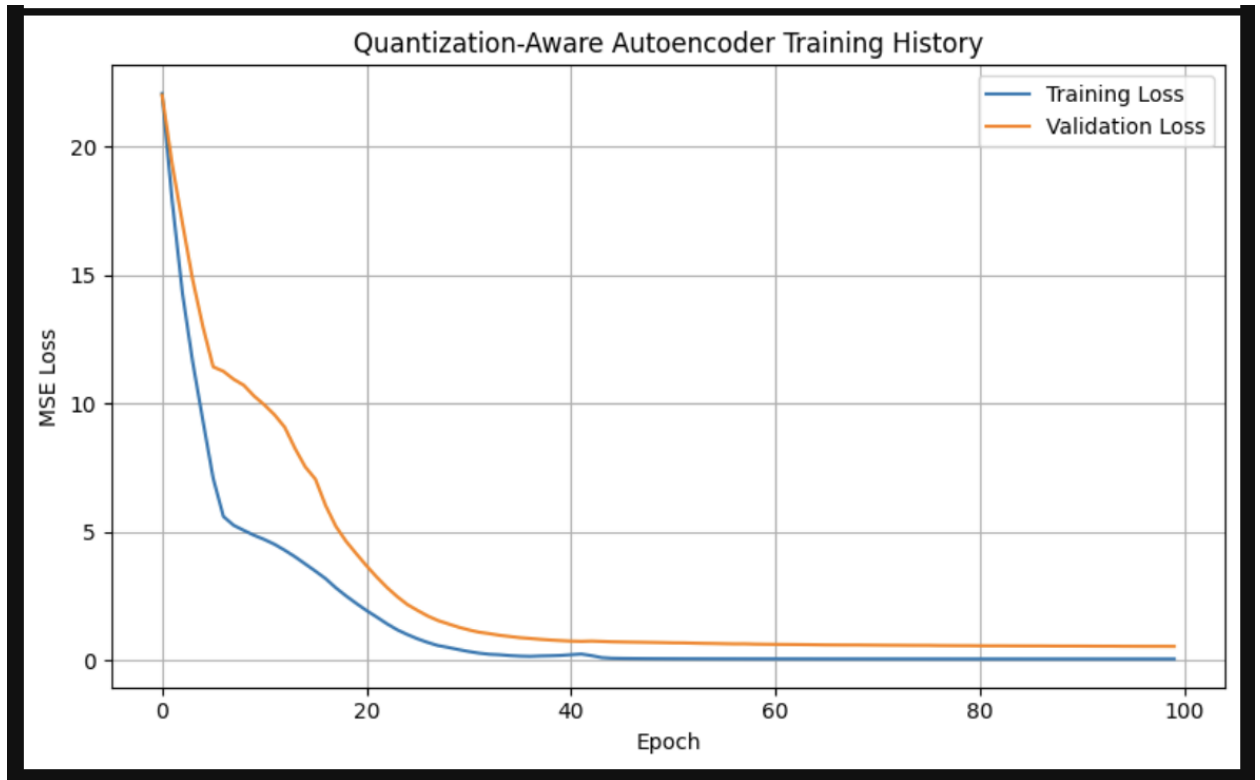
1. I chose the threshold value of .53 for the anomaly detection because it was the value I felt separates the normal and anomaly distributions .
2. Reconstruction error helps distinguish normal and anomalous activity by training the autoencoder during normal activity window. Normal activity will produce low reconstruction error because that is what the model will be taught during training. Meanwhile anomalous activity would produce a higher reconstruction error because that model has not learned that pattern.
3. The python notebook must preserve the trained int8 model, anomaly detection threshold, input dimensions, and the preprocessing steps.

4. It is important for the Arduino sketch to use the same preprocessing assumptions to maintain the predictions it is supposed to get. If not, it will be operating in a manner it was not trained to.
5. The main limitations of this anomaly detection method when deployed on a microcontroller is accuracy loss from the quantization. We might also run into a memory problem because the microcontroller is smaller.

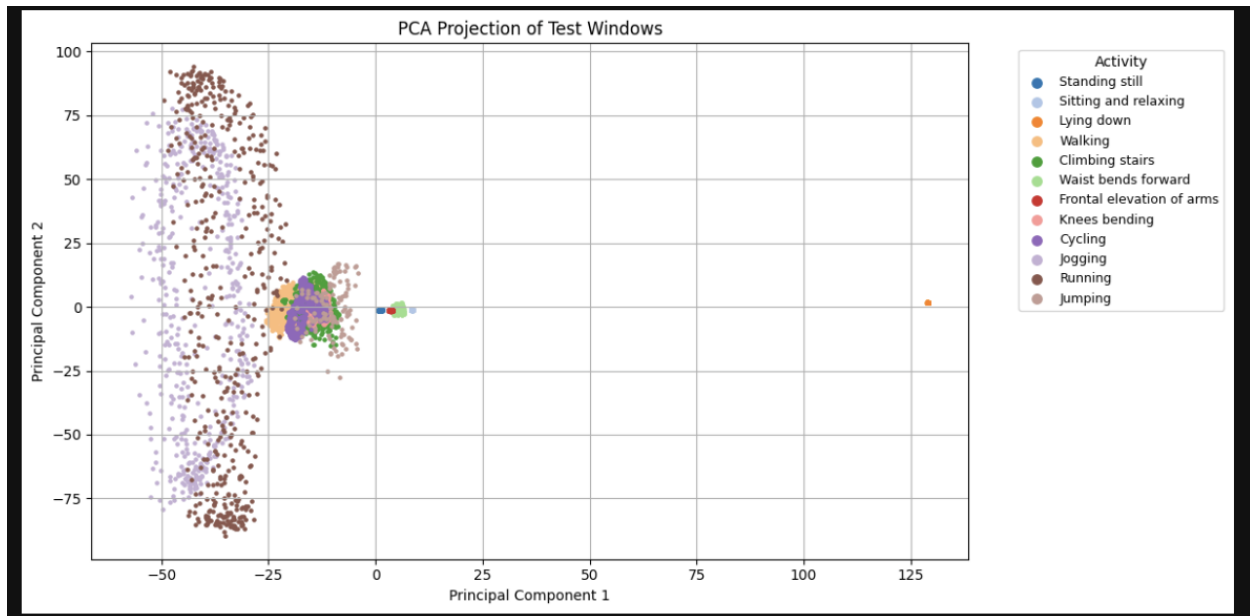
Part 2 screenshots:

- Training and validation loss curves:

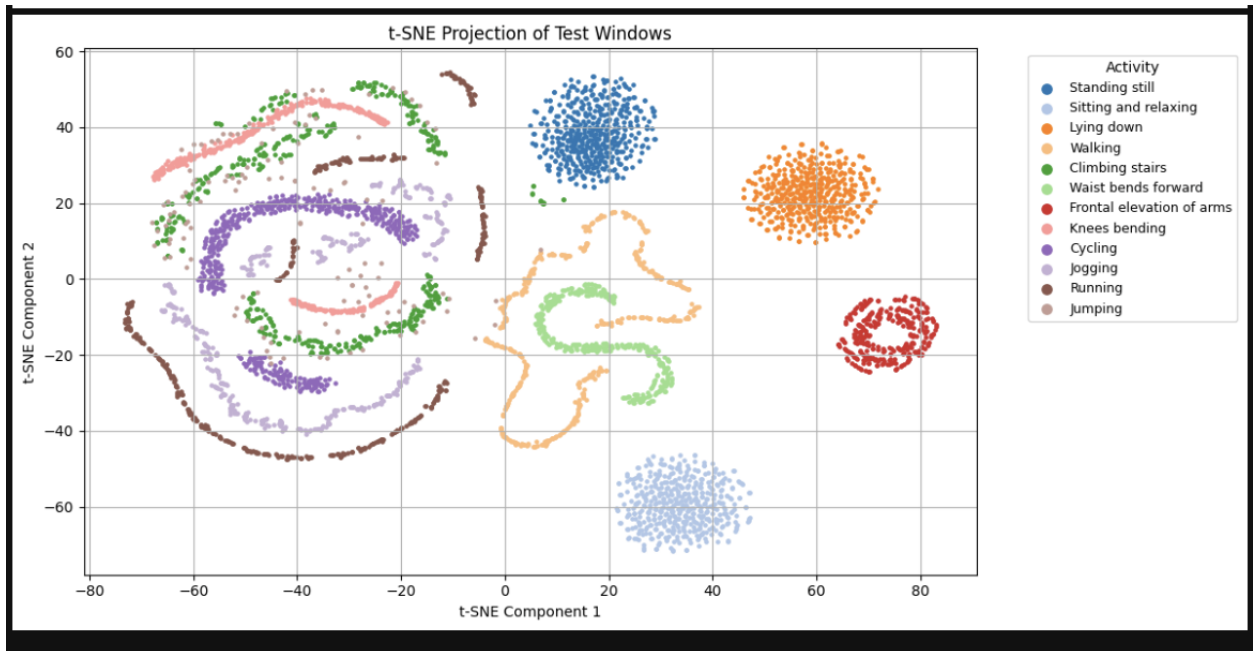




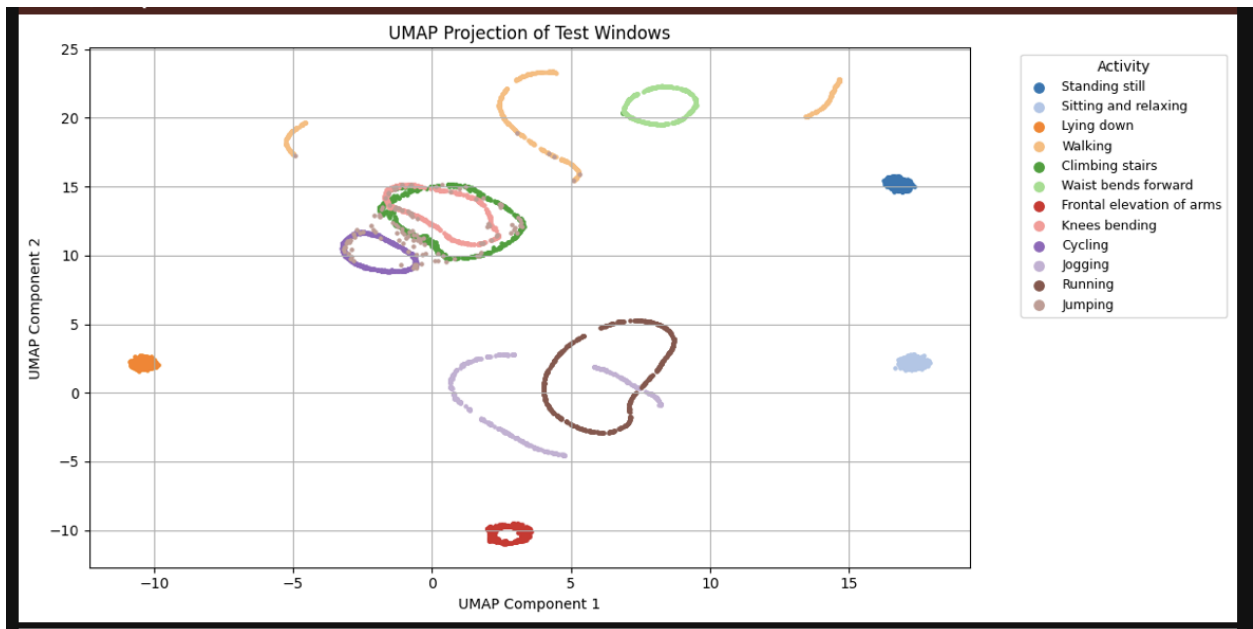
- PCA visualization :



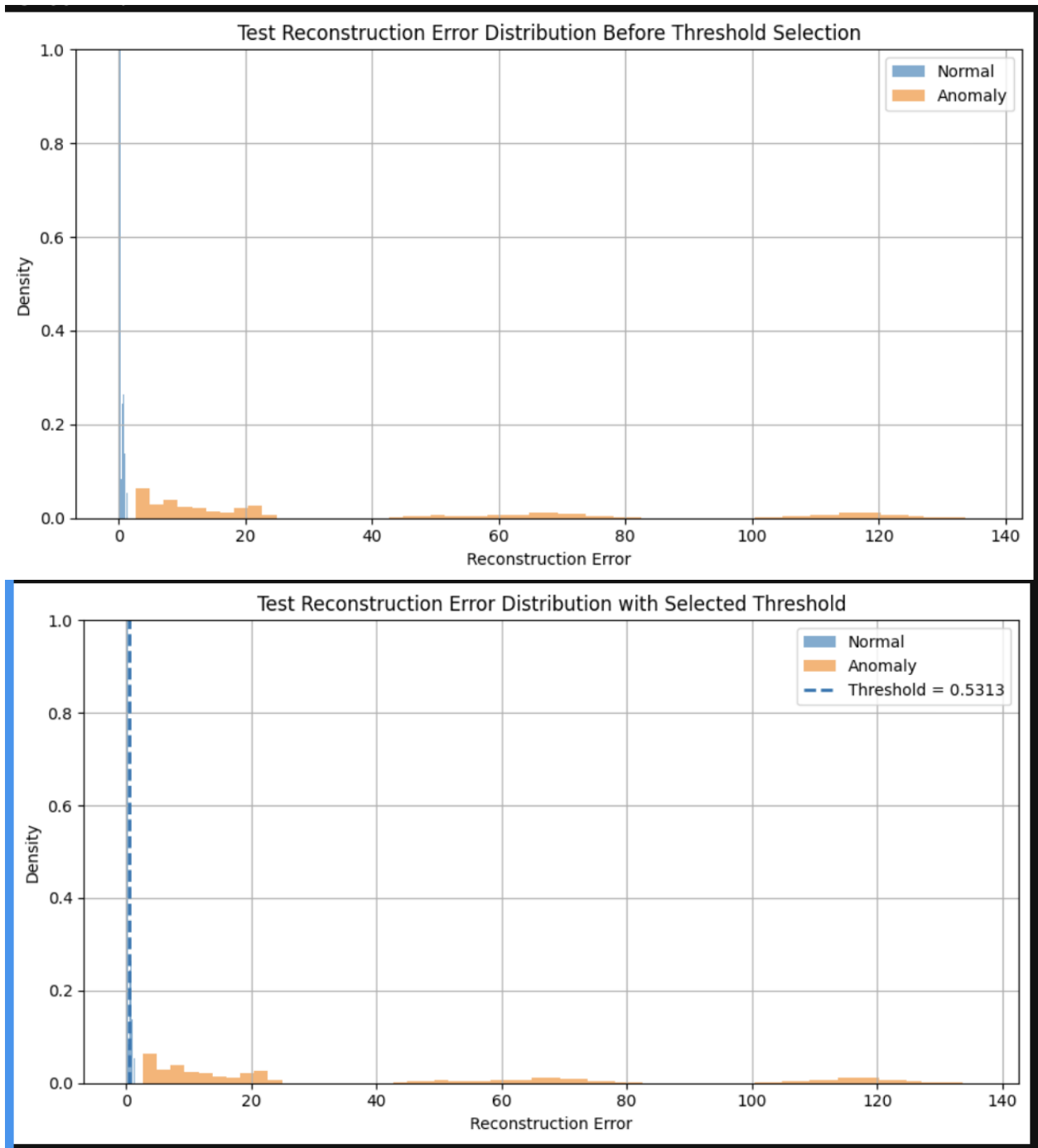
- t-SNE visualization :



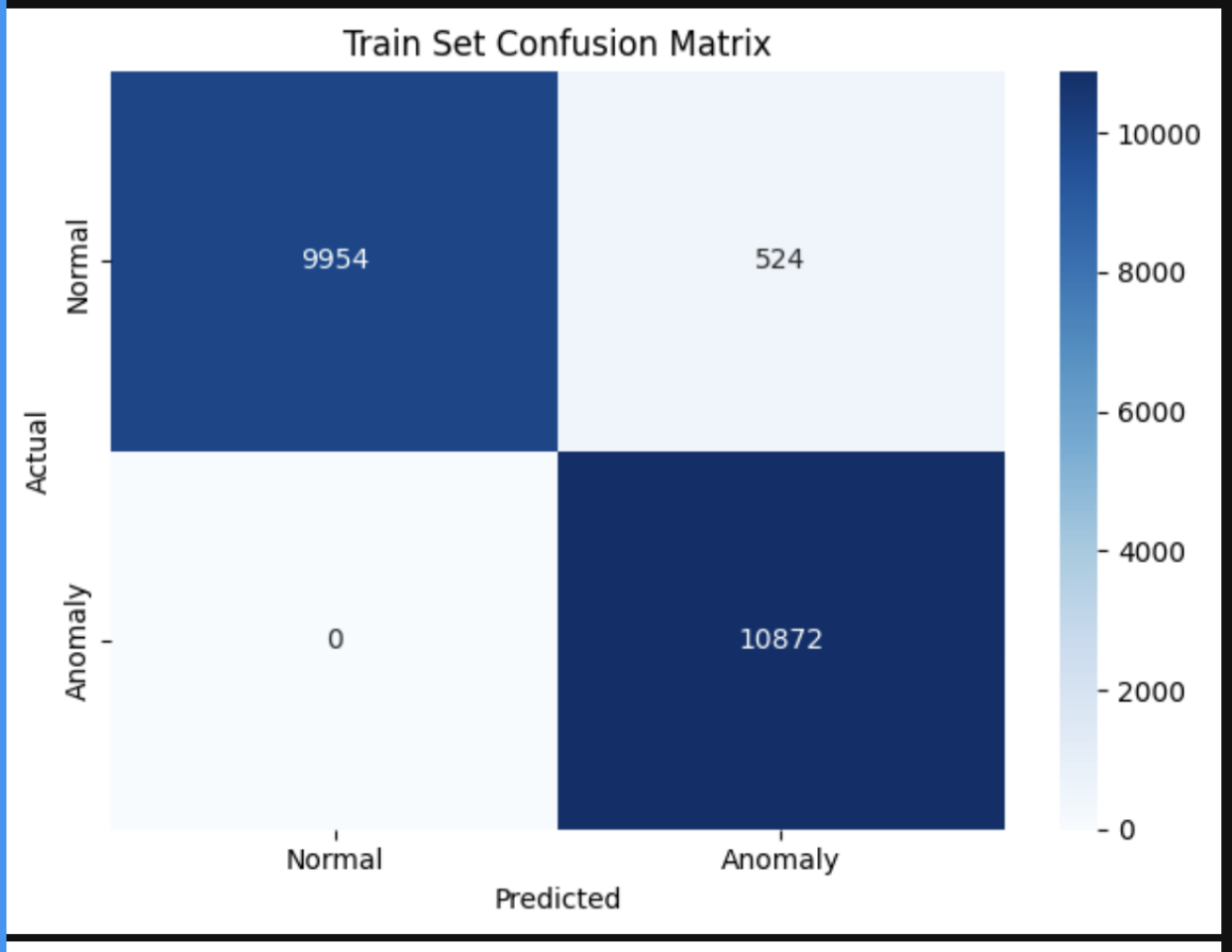
- UMAP visualization

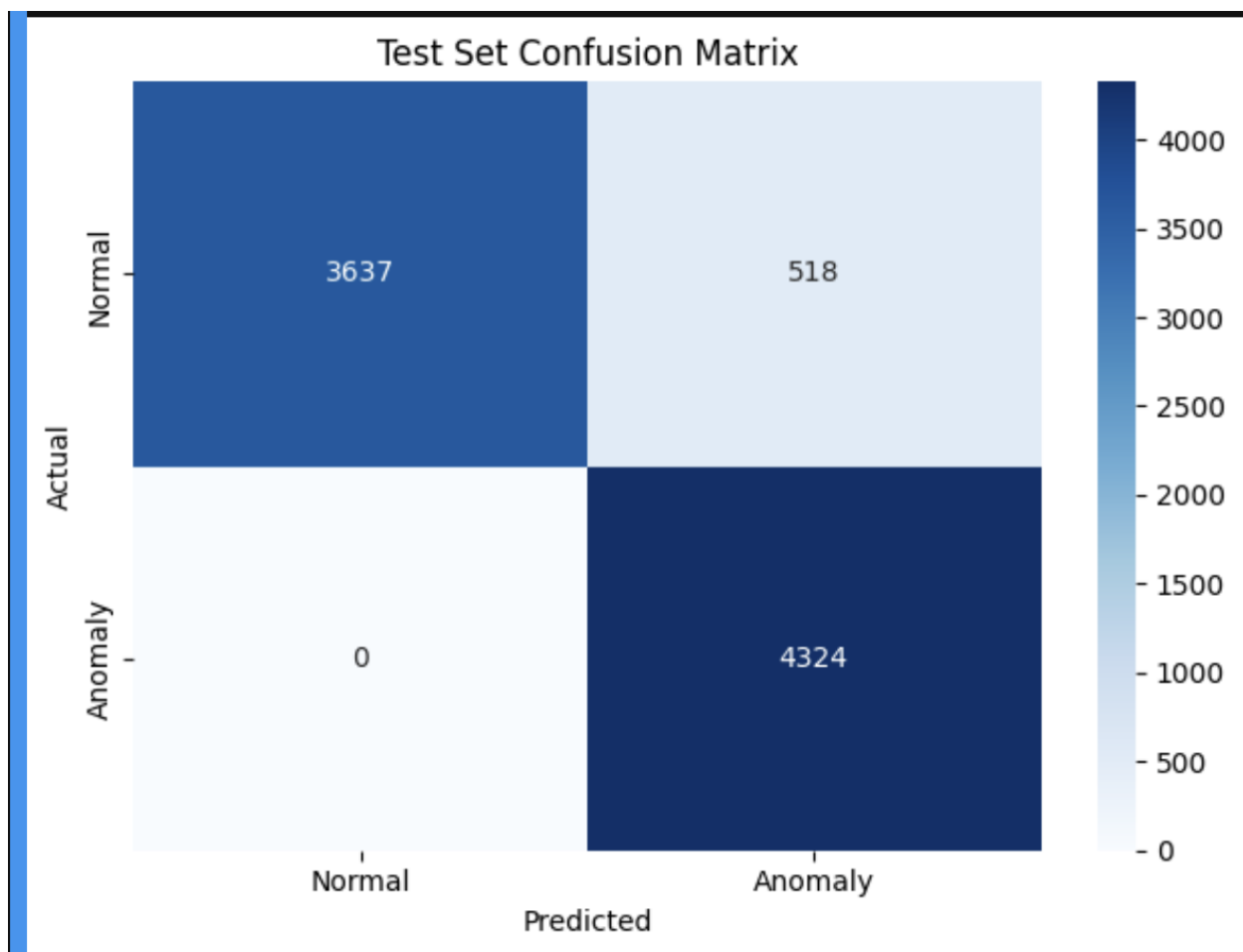


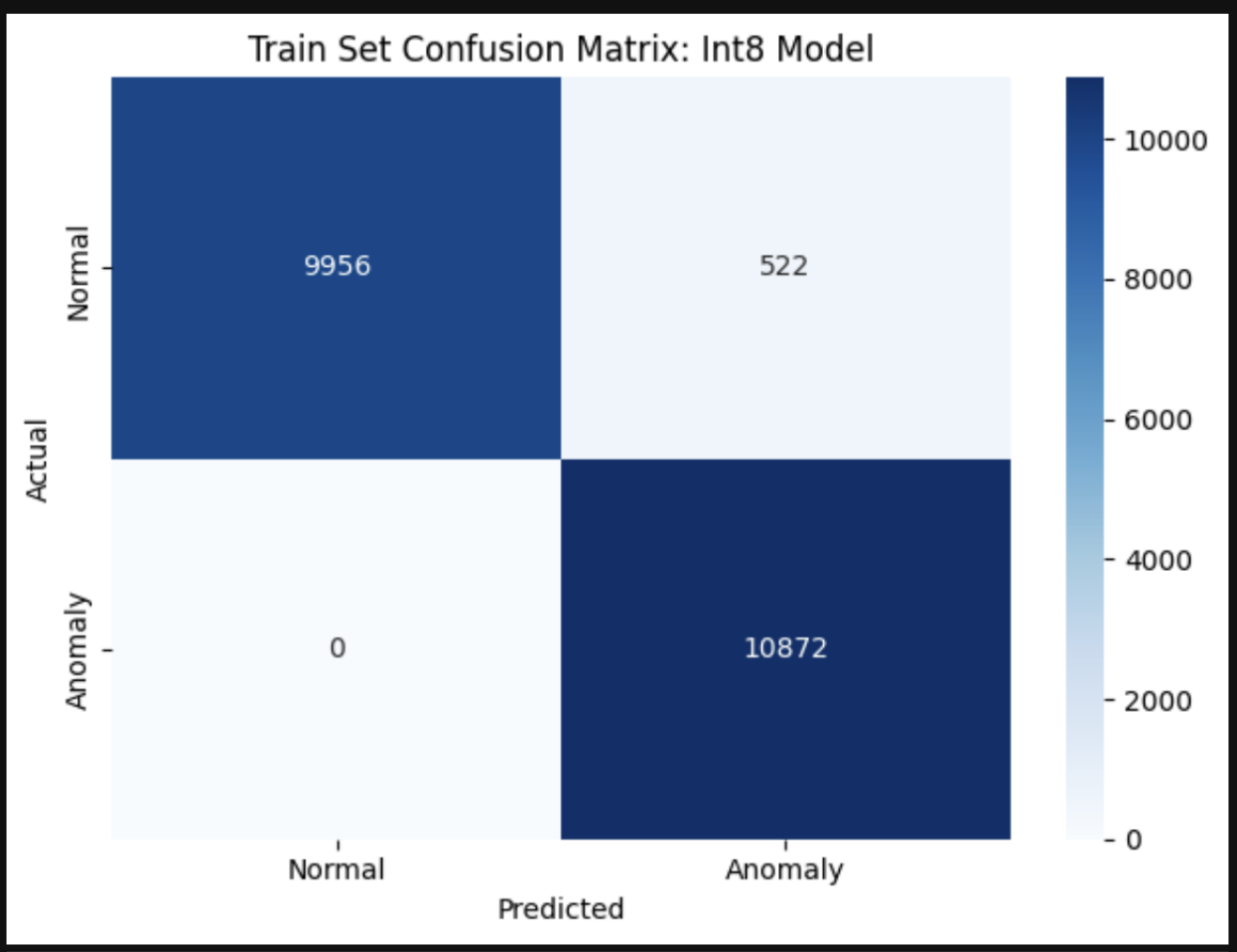
- Reconstruction error distribution :

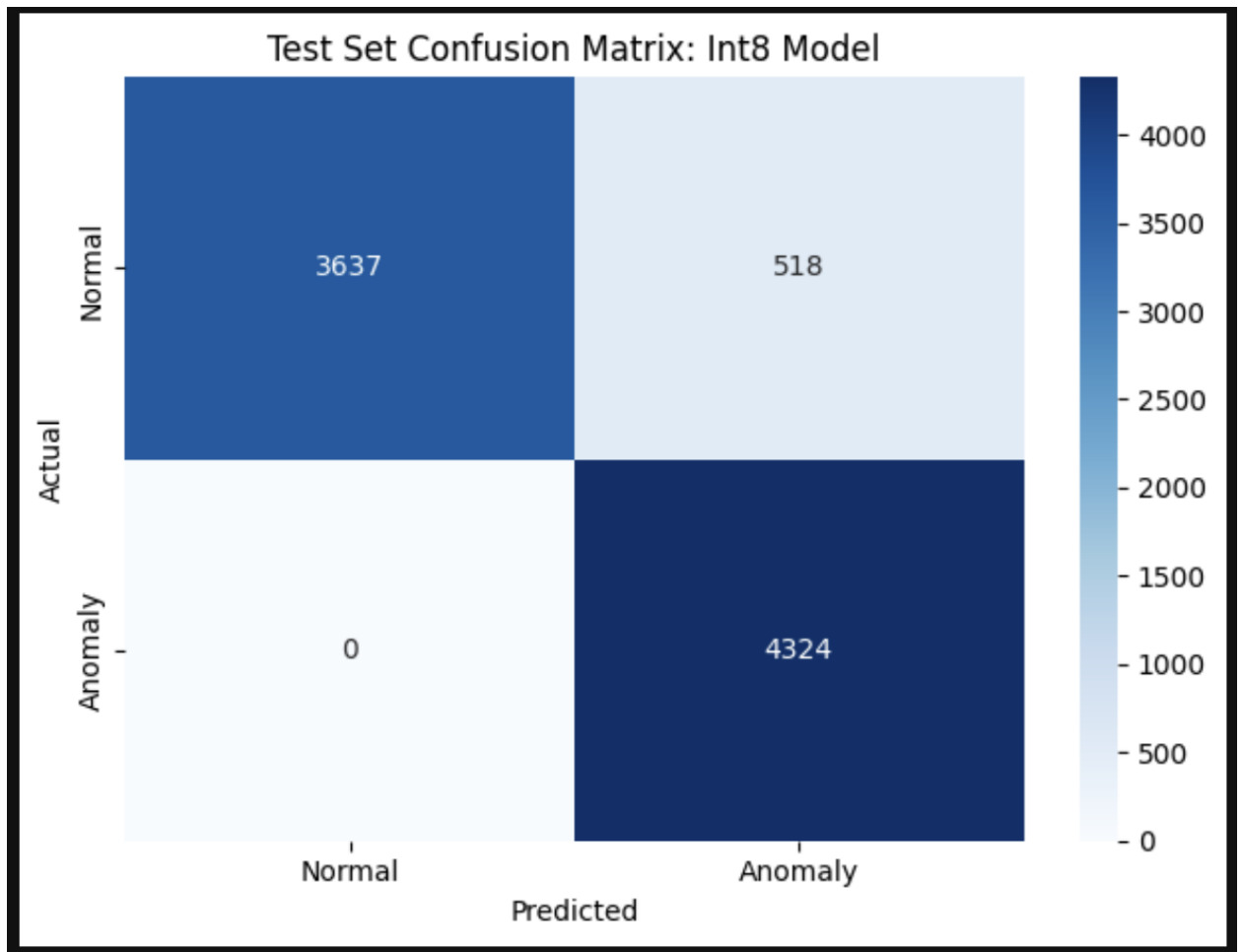


- Selected anomaly threshold: .5313
- Confusion Matrix :









- Classification report:

```
Input quantization: (0.08135765045881271, 18)
Output quantization: (0.07536008208990097, 10)
```

```
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
```

```
Train set evaluation: int8 model
```

	precision	recall	f1-score	support
Normal (0)	1.00	0.95	0.97	10478
Anomaly (1)	0.95	1.00	0.98	10872
accuracy			0.98	21350
macro avg	0.98	0.98	0.98	21350
weighted avg	0.98	0.98	0.98	21350

```
Test set evaluation: int8 model
```

	precision	recall	f1-score	support
Normal (0)	1.00	0.88	0.93	4155
Anomaly (1)	0.89	1.00	0.94	4324
accuracy			0.94	8479
macro avg	0.95	0.94	0.94	8479
weighted avg	0.95	0.94	0.94	8479

- Arduino monitor:

```
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